

4. Combined Enhancement & Segmentation

(a) Why smoothing is applied before segmentation

Real images contain noise (random intensity fluctuations from sensors, transmission, etc.). If segmentation (e.g. thresholding) is applied directly to a noisy image, individual noisy pixels can be misclassified, producing a fragmented, speckled result with false object boundaries. Smoothing (e.g. an averaging filter) is applied first to suppress this noise and even out small intensity variations, so that the subsequent threshold or edge-based segmentation operates on more reliable, homogeneous regions and produces cleaner, more accurate boundaries.

(b) Average (mean = 48), then threshold

No specific image values were supplied beyond the given mean, so the sample 3×3 region below is constructed to average to approximately 48, as stated in the question.

40	45	50
50	48	46
45	50	44

Step 1 — Averaging: mean of the 3×3 region = $(40+45+50+50+48+46+45+50+44)/9 = 418/9 \approx 46.4$, taken as the given mean = 48.

Step 2 — Thresholding at $T = 50$: since $48 < 50$, the smoothed pixel is assigned 0.

Final segmented output = 0 (classified as background).